Recognition, retrieval and confident errors

Multi-digit addition, a blind simulation, swapping memories, and a second task

Jeff Shrager

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Two theories, combined

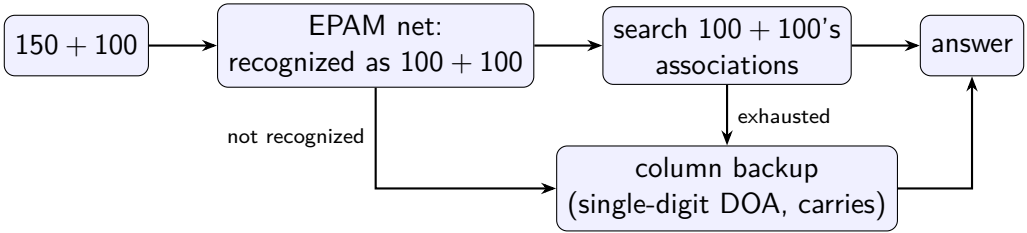
EPAM (Feigenbaum 1959)

- Stimuli are feature codes.
- A discrimination net sorts a code by testing features, down to a stored image.
- Learning grows a test only where two codes are confused, on the first difference noticed.
- So an unfamiliar code that agrees on every tested feature is *recognized as* a familiar one.

Distributions of associations (Siegler & Shrager 1984)

- Each problem has associations to answers, with strengths.
- Sample an answer in proportion to strength; state it if it clears a confidence criterion; try up to n times.
- Otherwise, use a backup strategy.
- Retrieval is used with probability $1 - (1 - R)^n$.

The multi-digit model



- Every postulate is a parameter: the encoding, the net, the strengths, the criterion, the search length, the backup.
- 18 predictions, each proved for every choice of them.

Confident errors

Let a familiar problem q have *sharp* associations (only its correct sum clears the criterion), and let x, with a different sum, be recognized as q. Then:

1. on q, whole-problem retrieval never errs;
2. on x, every whole-problem answer is wrong;
3. x uses the backup exactly as rarely as q does: $(1 - R_q)^n$.

In DOA alone, errors and backup use rise together. Confusion breaks the tie: a retrieved, wrong answer with no extra backup use.

A worked case

- Encoding: each digit in four bits (24 features). The net learns $100 + 100$ and $100 + 200$, which differ only in the second addend's hundreds digit, so that is all it tests.
- Associations: .90 on each problem's sum, .01 elsewhere; criterion .50; search length 1.
- $150 + 100$ is recognized as $100 + 100$.

$150 + 100$ answered 200, by retrieval	45/46
$150 + 100$ answered 250, by the columns	1/46

Checking the formalization: a blind simulation

The proofs show the predictions follow from the Lean definitions, not that the definitions say what the prose model says.

1. A prose specification, with no Lean names or predictions.
2. Predictions **preregistered** and committed before the simulator existed.
3. A fresh AI agent, given only the prose, wrote a Python simulator. Its tool use was audited.
4. A harness compared it with the preregistered predictions and with Lean's exact distributions.

Result: no failures. 243 outcome frequencies on 4 instances matched Lean's exact values, and every applicable prediction passed.

Memory as a parameter

The recognition step is now an interface: memory takes a probe and returns a *distribution* over stored items (or nothing).

- EPAM's net is one memory: deterministic.
- ACT-R's declarative memory is another: base-level activation that decays, partial matching, a retrieval threshold.
- With decay 1 and temperature 1, ACT-R's retrieval probabilities are exactly rational, so Lean computes them exactly.

Each prediction now states the law of memory it needs. Swap the memory and keep every theorem whose laws still hold.

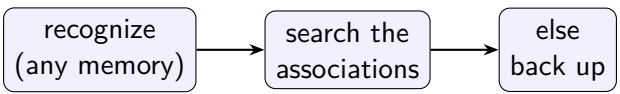
The same problems, two memories

	EPAM	ACT-R
$150 + 100$ answered 200 (the neighbour's sum)	45/46	225/529
$150 + 100$ answered 300 (differs on a tested digit)	0	225/1058
$150 + 100$ answered correctly	1/46	383/1058
errors on the practiced $100 + 100$	0	225/782
the same, after also studying $100 + 300$	0	3375/5842

ACT-R: confusion is graded, nothing is retrieved for certain, and new practice interferes with old.

The composition, extracted

The model's trial is one case of a general composition, now in the shared library:



- Over any problems and any answers, with "the correct answer" a parameter.
- Confident errors, and the rest, are proved once for the composition.
- Multi-digit addition is an instance: one bridge lemma carries every probability across, and its theorems keep their names.

A second task: naming people

Faces are codes; familiar Ann (000) and Bob (110); stranger Carl (100), one feature from each. The backup is "don't know".

	EPAM	ACT-R
Carl named Bob	45/46	225/391
Carl named Ann	0	225/782
Carl: "don't know"	1/46	107/782
Bob: "don't know"	1/46	229/2254
Bob named Ann	0	225/2254

With EPAM, *confident misnaming* is inherited from the composition's theorem, for every search length. No arithmetic anywhere.

From laws: which memories give confident errors?

Theorem. Any memory with three laws (*determinacy*, *certainty*, *stability*) produces confident errors whenever a stranger can be taken for a studied item with sharp associations.

- In the matrix of memory families, only discrimination (EPAM) has all three.
- Face naming's confident misnaming is derived this way, from EPAM's matrix cells plus one computed fact (Carl can be taken for Bob).
- The evidence judges that people lack certainty. So explaining confident errors in people needs a graded version: these laws with high probability, not probability 1.

Limits, and next

- **No timing.** "Retrieved, with no extra backup use" is what is proved; "fast" needs a timing model.
- **One trial at a time.** Learning across trials (feedback, discrimination, strengthening) makes memory states random; the predicted sequence (confident error, then column work, then fluent retrieval) is the next target.
- **Formal fidelity.** ACT-R here is one restricted formalization; checking it against PyACTUp, the community implementation, is planned.
- **A graded confident-errors theorem**, for memories without certainty.